

2.29 Numerical Fluid Mechanics

Spring 2015 – Lecture 22

REVIEW Lecture 21:

- Time-Marching Methods and ODEs – IVPs: End
 - Multistep/Multipoint Methods: Adams Methods
 - Additional points are at past time steps
 - Practical CFD Methods
 - Implicit Nonlinear systems
 - Deferred-correction Approach
- Complex Geometries
 - Different types of grids
 - Choice of variable arrangements:
- Grid Generation
 - Basic concepts and structured grids
 - Stretched grids
 - Algebraic methods (for stretched grids), Transfinite Interpolation

TODAY (Lecture 22):

Grid Generation and Intro. to Finite Elements

- Grid Generation
 - Basic concepts and structured grids, cont'd
 - General coordinate transformation
 - Differential equation methods
 - Conformal mapping methods
 - Unstructured grid generation
 - Delaunay Triangulation
 - Advancing Front method
- Finite Element Methods
 - Introduction
 - Method of Weighted Residuals: Galerkin, Subdomain and Collocation
 - General Approach to Finite Elements:
 - Steps in setting-up and solving the discrete FE system
 - Galerkin Examples in 1D and 2D
 - Computational Galerkin Methods for PDE: general case
 - Variations of MWR: summary
 - Finite Elements and their basis functions on local coordinates (1D and 2D)

References and Reading Assignments

Complex Geometries and Grid Generation

- Chapter 8 on “Complex Geometries” of “J. H. Ferziger and M. Peric, Computational Methods for Fluid Dynamics. Springer, NY, 3rd edition, 2002”
- Chapter 9 on “Grid Generation” of T. Cebeci, J. P. Shao, F. Kafyeke and E. Laurendeau, *Computational Fluid Dynamics for Engineers*. Springer, 2005.
- Chapter 13 on “Grid Generation” of Fletcher, Computational Techniques for Fluid Dynamics. Springer, 2003.
- Ref on Grid Generation only:
 - Thompson, J.F., Warsi Z.U.A. and C.W. Mastin, “Numerical Grid Generation, Foundations and Applications”, North Holland, 1985

Grid Generation for Structured Grids:

General Coordinate transformation

- For structured grids, mapping of coordinates from Cartesian domain to physical domain is defined by a transformation: $x_i = x_i(\xi_j)$ ($i \& j = 1, 2, 3$)

$$J = \det \left(\frac{\partial x_i}{\partial \xi_j} \right) = \begin{vmatrix} \frac{\partial x_1}{\partial \xi_1} & \frac{\partial x_1}{\partial \xi_2} & \frac{\partial x_1}{\partial \xi_3} \\ \frac{\partial x_2}{\partial \xi_1} & \frac{\partial x_2}{\partial \xi_2} & \frac{\partial x_2}{\partial \xi_3} \\ \frac{\partial x_3}{\partial \xi_1} & \frac{\partial x_3}{\partial \xi_2} & \frac{\partial x_3}{\partial \xi_3} \end{vmatrix}$$

- All transformations are characterized by their Jacobian determinant J .

- For Cartesian vector components, one only needs to transform derivatives. One has:

$$\frac{\partial \phi}{\partial x_i} = \frac{\partial \phi}{\partial \xi_j} \frac{\partial \xi_j}{\partial x_i} = \frac{\partial \phi}{\partial \xi_j} \frac{\beta^{ij}}{J}, \quad \text{where } \beta^{ij} \text{ represents the cofactor of } \frac{\partial x_i}{\partial \xi_j} \text{ (element } i, j \text{ of Jacobian matrix)}$$

- In 2D, $x = x(\xi, \eta)$ and $\phi = \phi(\xi, \eta)$, this leads to:

$$\frac{\partial \phi}{\partial x} = \frac{\partial \phi}{\partial \xi} \frac{\partial \xi}{\partial x} + \frac{\partial \phi}{\partial \eta} \frac{\partial \eta}{\partial x} = \frac{\partial \phi}{\partial \xi} \frac{\beta^{11}}{J} + \frac{\partial \phi}{\partial \eta} \frac{\beta^{12}}{J} = \frac{1}{J} \left(\frac{\partial \phi}{\partial \xi} \frac{\partial y}{\partial \eta} - \frac{\partial \phi}{\partial \eta} \frac{\partial y}{\partial \xi} \right)$$

Recall: the minor element m_{ij} corresponding to a_{ij} is the determinant of the submatrix that remains after the i^{th} row and the j^{th} column are deleted from $\mathbf{A}$. The cofactor c_{ij} of a_{ij} is: $c_{ij} = (-1)^{i+j} m_{ij}$

Grid Generation for Structured Grids: General Coordinate transformation, Cont'd

- How do the conservation equations transform?

The generic conservation equation in Cartesian coordinates:

$$\frac{\partial \rho \phi}{\partial t} + \nabla \cdot (\rho \phi \bar{v}) = \nabla \cdot (k \nabla \phi) + s_\phi \quad \Leftrightarrow \quad \frac{\partial \rho \phi}{\partial t} + \frac{\partial}{\partial x_j} \left(\underbrace{\rho \phi v_j}_{\text{advection}} - \underbrace{k \frac{\partial \phi}{\partial x_j}}_{\text{diffusion}} \right) = s_\phi$$

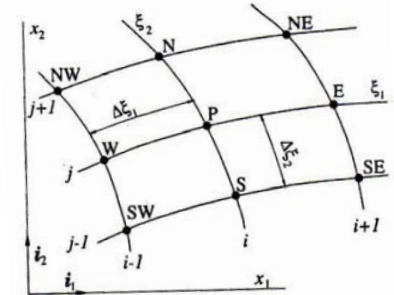
- becomes:

$$J \frac{\partial \rho \phi}{\partial t} + \frac{\partial}{\partial \xi_j} \left(\rho \phi U_j - \frac{k}{J} \left(\frac{\partial \phi}{\partial \xi_m} B^{mj} \right) \right) = J s_\phi$$

where:

$$\left\{ \begin{array}{l} U_j = v_k \beta^{kj} = v_1 \beta^{1j} + v_2 \beta^{2j} + v_3 \beta^{3j} \quad \text{is proportional to the velocity component aligned with } \vec{\xi}_j \\ \hspace{10em} (\text{normal to } \xi_j = \text{const.}) \\ B^{mj} = \beta^{kj} \beta^{km} = \beta^{1j} \beta^{1m} + \beta^{2j} \beta^{2m} + \beta^{3j} \beta^{3m} \quad \text{are coefficients, sum of products of cofactors } \beta^{ij} \end{array} \right.$$

- As a result, each 1st derivative term is replaced by a sum of three terms which contains derivatives of the coordinates as coefficients
- Unusual features of conservation equations in non-orthogonal grids:
 - Mixed derivatives appear in the diffusive terms and metrics coefficients appear in the continuity eqn.



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Structured Grids: Gen. Coord. transformation, Cont'd

Some Comments

- Coordinate transformation often presented only as a means of converting a complicated non-orthogonal grid into a simple, uniform Cartesian grid (the computational domain, whose grid-spacing is arbitrary)
- However, simplification is only apparent:
 - Yes, the computational grid is simpler than the original physical one
 - But, the information about the **complexity** in the computational domain **is now in the metric coefficients of the transformed equations**
 - i.e. discretization of computational domain is now simple, but the calculation of the Jacobian and other geometric information is not trivial (the difficulty is hidden in the metric coefficients)
- As mentioned earlier, FD method can in principle be applied to unstructured grids: specify a local shape function, differentiate and write FD equations. Has not yet been done.

Grid Generation for Structured Grids: Differential Equation Methods

- Grid transformation relations determined by a finite-difference solution of PDEs
 - For 2D problems, two elliptic (Poisson) PDEs are solved
 - Can be done for any coordinate systems, but here we will use Cartesian coordinates. The 2D transformation is then:
 - From the physical domain (x, y) to the computational domain (ξ, η)
 - At physical boundaries, one of ξ, η is constant, the other is monotonically varying
 - At interior points:

$$\begin{aligned}\frac{\partial^2 \xi}{\partial x^2} + \frac{\partial^2 \xi}{\partial y^2} &= P(\xi, \eta) \\ \frac{\partial^2 \eta}{\partial x^2} + \frac{\partial^2 \eta}{\partial y^2} &= Q(\xi, \eta)\end{aligned}$$

where $P(\xi, \eta)$ and $Q(\xi, \eta)$ are called the “control functions”

- Their selection allows to concentrate the ξ, η lines in specific regions
- If they are null, coordinates will tend to be equally spaced away from boundaries
- Boundary conditions: ξ, η specified on boundaries of physical domain

Grid Generation for Structured Grids: Differential Equation Methods, Cont'd

- Computations to generate the grid mapping are actually carried out in the computational domain (ξ, η) itself !
 - don't want to solve the elliptic problem in the complex physical domain!
- Using the general rule, the elliptic problem is transformed into:

$$\begin{aligned}\alpha \frac{\partial^2 x}{\partial \xi^2} - 2\beta \frac{\partial^2 x}{\partial \xi \partial \eta} + \gamma \frac{\partial^2 x}{\partial \eta^2} + J^2 \left(P \frac{\partial x}{\partial \xi} + Q \frac{\partial x}{\partial \eta} \right) &= 0 \\ \alpha \frac{\partial^2 y}{\partial \xi^2} - 2\beta \frac{\partial^2 y}{\partial \xi \partial \eta} + \gamma \frac{\partial^2 y}{\partial \eta^2} + J^2 \left(P \frac{\partial y}{\partial \xi} + Q \frac{\partial y}{\partial \eta} \right) &= 0\end{aligned}$$

where $\alpha = x_\eta^2 + y_\eta^2$; $\beta = x_\xi x_\eta + y_\xi y_\eta$; $\gamma = x_\xi^2 + y_\xi^2$; $J = x_\xi y_\eta - x_\eta y_\xi$ (with $x_\xi = \frac{\partial x}{\partial \xi}$, etc)

- Boundary conditions are now the transformed values of the BCs in (x, y) domain: they are the values of the positions (x, y) of the grid points on the physical domain mapped to their locations in the computational domain
- Equations can be solved by FD method to determine values of every grid point (x, y) in the interior of the physical domain
- Method developed by Thomson et al., 1985 (see ref)

Grid Generation for Structured Grids: Differential Equation Methods, Example

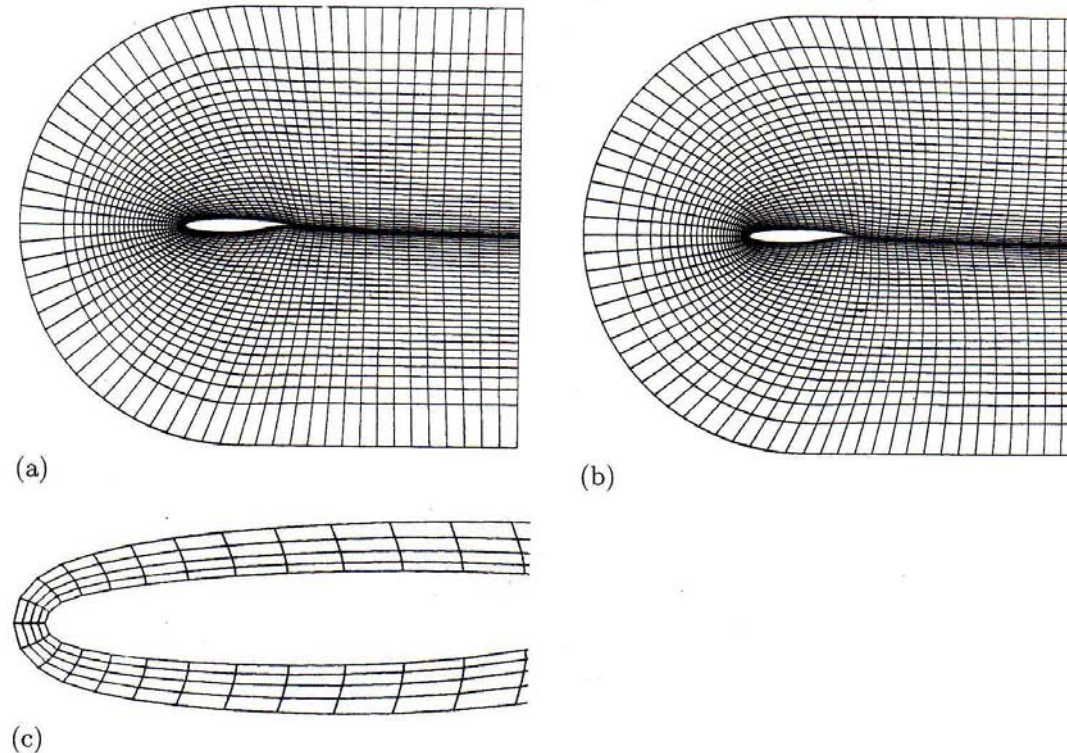


Fig. 9.13. (a) Starting algebraic C-grid around an airfoil section; 70×30 grid points; inner spacing $\Delta S_1 = 0.015c$, outer spacing $\Delta S_2 = 0.3c$, (b) Elliptic C-grid obtained after smoothing the algebraic grid of (a) by the solution of Poisson equations (50 iterations), (c) Close-up of the C-grid showing the application of orthogonality conditions near the leading edge region.

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Grid Generation for Structured Grids: Conformal Mapping Methods

- Conformal mapping schemes are analytical or partially analytical (as opposed to differential equation methods)
- Restricted to two dimensional flows (based on complex variables): useful for airfoils
- Examples:

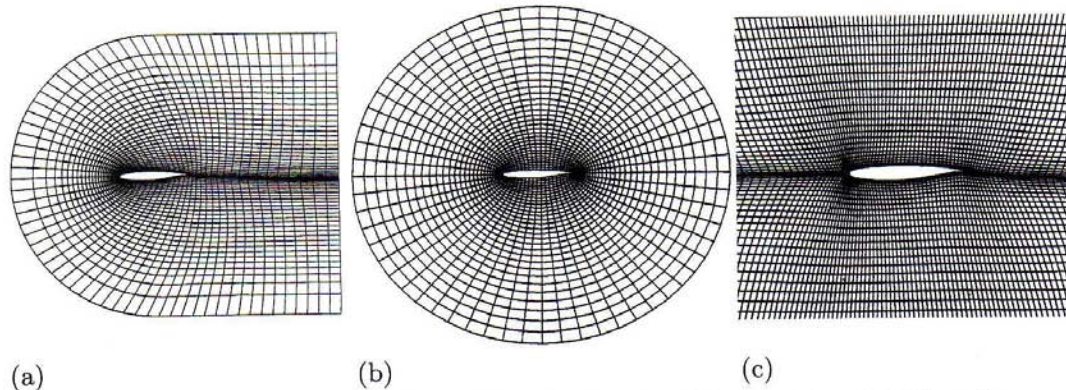


Fig. 9.14. Three common grids for airfoils. (a) C-grid, (b) O-grid, and (c) H-grid.

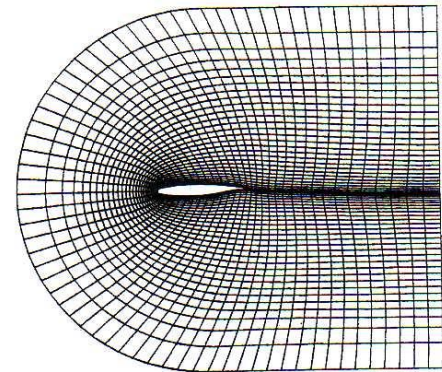
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- C-mesh: high density near leading edge of airfoil and good wake
- O-mesh: high density near leading and trailing edge of airfoil
- H-mesh: two sets of mesh lines similar to a Cartesian mesh, which is easiest to generate. Its mesh lines are often well aligned with streamlines

Grid Generation for Structured Grids: Conformal Mapping Methods: Example

- C-mesh example is generated by a parabolic mapping function
- It is essentially a set of confocal, orthogonal parabolas wrapping around the airfoil

- The mapping is defined by:
$$2(x + iy) = (\xi + i\eta)^2$$
or
$$2x = \xi^2 - \eta^2 ; \quad y = \xi\eta$$



- Inverse transformation:

$$\xi^2 = \sqrt{x^2 + y^2} + x ; \quad \eta^2 = \sqrt{x^2 + y^2} - x$$

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- Polar coordinates can be used for easier “physical plane” to “computational plane” transformation.
- In conformal mapping, singular point is point where mapping fails (here, it is the origin) => move it to half the distance from the nose radius

Grid Generation: Unstructured Grids

- Generating unstructured grid is complicated but now relatively automated in “classic” cases
- Involves succession of smoothing techniques that attempt to align elements with boundaries of physical domain
- Decompose domain into blocks to decouple the problems
- Need to define point positions and connections
- Most popular algorithms:
 - Delaunay Triangulation Method
 - Advancing Front Method
- Two schools of thought: structured vs. unstructured, what is best for CFD?

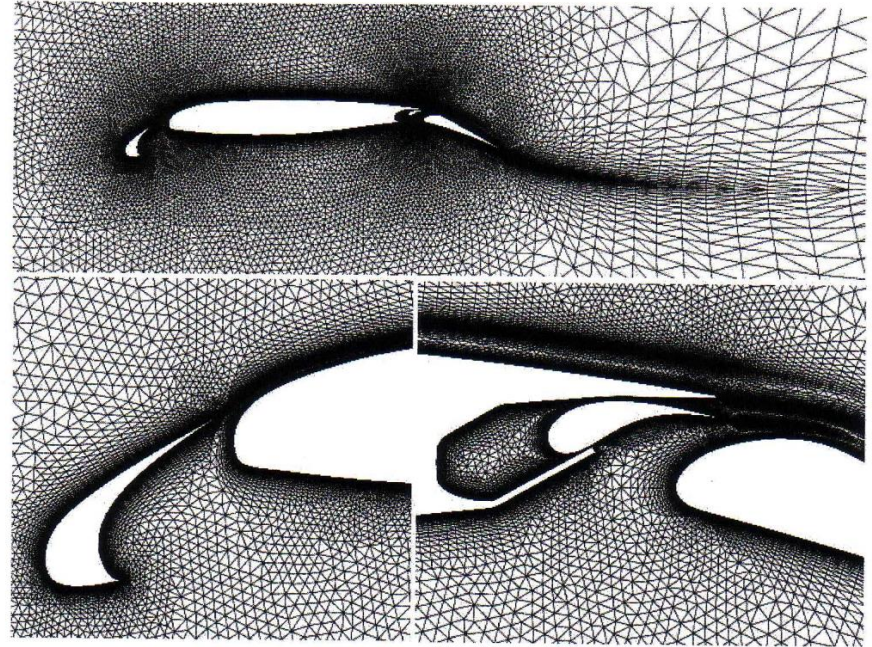


Fig. 9.16. 2D Unstructured grid for Navier–Stokes computations of a multi-element airfoil generated with the hybrid advancing front Delaunay method of Mavriplis [6].

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- Structured grids: simpler grid and straightforward treatment of algebraic system, but mesh generation constraints on complex geometries
- Unstructured grids: generated faster on complex domains, easier mesh refinements, but data storage and solution of algebraic system more complex

Grid Generation: Unstructured Grids

ay Triangulation (DT)

- Use a simple criterion to connect points to form conforming, non-intersecting elements
- Maximizes minimum angle in each triangle
- Not unique
- Task of point generation is done independently of connection generation
- Based on Dirichlet's domain decomposition into a set of packed convex regions:
 - For a given set of points P , the space is subdivided into regions in such a way that each region is the space closer to P than to any other point = Dirichlet tessellation

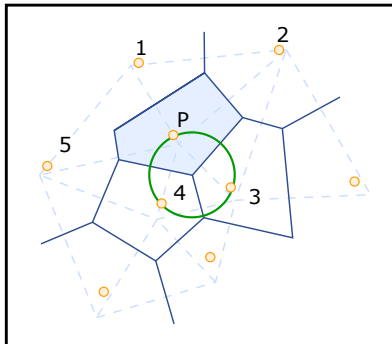


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Note: at the end, points P are at summits of triangles

- This geometrical construction is known as the **Dirichlet (Voronoi) tessellation**
- The tessellation of a closed domain results in a set of non-overlapping convex regions called **Voronoi regions/polygons**
- The sides of the polygon around P is made of segments bisectors of lines joining P to its neighbors: if all pair of such P points with a common segment are joined by straight lines, the result is a Delaunay Triangulation
- Each vortex of a Voronoi diagram is then the circumcenter of the triangle formed by the three points of a Delaunay triangle
- Criterion: the circumcircle can not contain any other point than these three points

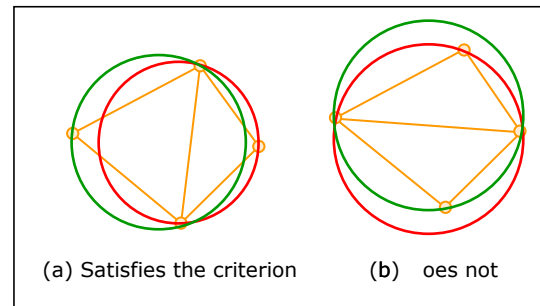


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Grid Generation: Unstructured Grids

- **Advancing Front Method**

- In this method, the tetrahedras are built progressively, inward from the boundary
- An active front is maintained where new tetrahedra are formed
- For each triangle on the edge of the front, an ideal location for a new third node is computed
- Requires intersection checks to ensure triangles don't overlap

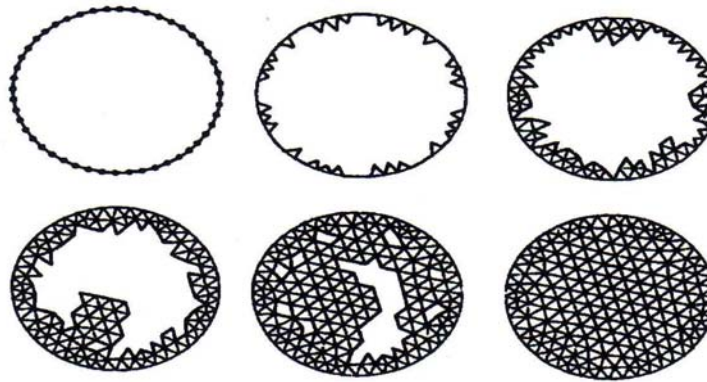


Fig. 9.20. Advancing Front technique for unstructured grid generation.

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- In 3D, the Delaunay Triangulation is preferred (faster)

References and Reading Assignments

Finite Element Methods

- Chapters 31 on “Finite Elements” of “Chapra and Canale, Numerical Methods for Engineers, 2006.”
- Lapidus and Pinder, 1982: Numerical solutions of PDEs in Science and Engineering.
- Chapter 5 on “Weighted Residuals Methods” of Fletcher, Computational Techniques for Fluid Dynamics. Springer, 2003.
- Some Refs on Finite Elements only:
 - Hesthaven J.S. and T. Warburton. Nodal discontinuous Galerkin methods, vol. 54 of Texts in Applied Mathematics. Springer, New York, 2008. Algorithms, analysis, and applications
 - Mathematical aspects of discontinuous Galerkin methods (Di Pietro and Ern, 2012)
 - Theory and Practice of Finite Elements (Ern and Guermond, 2004)

FINITE ELEMENT METHODS: Introduction

- Finite Difference Methods: based on a discretization of the differential form of the conservation equations
 - Solution domain divided in a grid of discrete points or nodes
 - PDE replaced by finite-divided differences = “point-wise” approximation
 - Harder to apply to complex geometries
- Finite Volume Methods: based on a discretization of the integral forms of the conservation equations:
 - Grid generation: divide domain into set of discrete control volumes (CVs)
 - Discretize integral equation
 - Solve the resultant discrete volume/flux equations
- Finite Element Methods: based on reformulation of PDEs into minimization problem, pre-assuming piecewise shape of solution over finite elements
 - Grid generation: divide the domain into simply shaped regions or “elements”
 - Develop approximate solution of the PDE for each of these elements
 - Link together or assemble these individual element solutions, ensuring some continuity at inter-element boundaries => PDE is satisfied in piecewise fashion

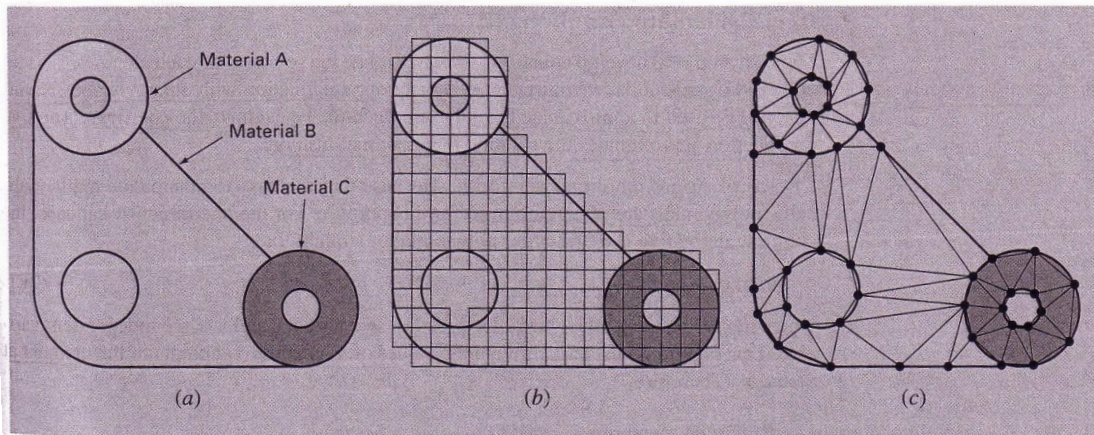
Finite Elements: Introduction, Cont'd

ginally based on the Direct Stiffness Method (Navier in 1826) and Rayleigh-Ritz, and further developed in its current form in the 1950's (Turner and others)

- Can replace somewhat “ad-hoc” integrations of FV with more rigorous minimization principles
- Originally more difficulties with convection-dominated (fluid) problems, applied to solids with diffusion-dominated properties

Comparison of FD and FE grids

[a] A gasket with irregular geometry and nonhomogeneous composition. [b] Such a system is very difficult to model with a finite-difference approach. This is due to the fact that complicated approximations are required at the boundaries of the system and at the boundaries between regions of differing composition. [c] A finite-element discretization is much better suited for such systems.



Examples of Finite elements

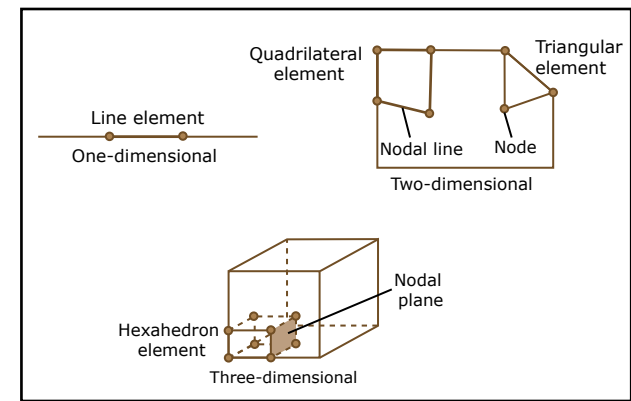


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Finite Elements: Introduction, Cont'd

- Classic example: Rayleigh-Ritz / Calculus of variations

- Finding the solution of $\frac{\partial^2 u}{\partial x^2} = -f \quad \text{on }]0,1[$

is the same as finding u that minimizes $J(u) = \int_0^1 \frac{1}{2} \left(\frac{\partial u}{\partial x} \right)^2 - u f \, dx$

- R-R approximation:

- Expand unknown u into shape/trial functions

$$u(x) = \sum_{i=1}^n a_i \phi_i(x)$$

and find coefficients a_i such that $J(u)$ is minimized

- Finite Elements:

- As Rayleigh-Ritz but choose trial functions to be piecewise shape function defined over set of elements, with some continuity across elements

Finite Elements: Introduction, Cont'd

Method of Weighted Residuals

- There are several avenues that lead to the same FE formulation
 - A conceptually simple, yet mathematically rigorous, approach is the Method of Weighted Residuals (MWR)
 - Two special cases of MWR: the Galerkin and Collocation Methods
- In the MWR, the desired function u is replaced by a finite series approximation into shape/basis/interpolation functions:

$$\tilde{u}(x) = \sum_{i=1}^n a_i \phi_i(x)$$

- $\phi_i(x)$ chosen such they satisfy the boundary conditions of the problem
- But, they will not in general satisfy the PDE: $L(u) = f$
 $\Rightarrow$ they lead to a residual: $L(\tilde{u}(x)) - f(x) = R(x) \neq 0$
- The objective is to select the undetermined coefficients a_i so that this residual is minimized in some sense

Finite Elements:

Method of Weigthed Residuals, Cont'd

- One possible choice is to set the integral of the residual to be zero. This only leads to one equation for n unknowns

⇒ Introduce the so-called weighting functions $w_i(x)$ $i=1,2,\dots,n$, and set the integral of each of the weighted residuals to zero to yield n independent equations:

$$\int_t^L \int R(x) w_i(x) dx dt = 0, \quad i = 1, 2, \dots, n$$

- In 3D, this becomes:

$$\int_t \int_V R(\mathbf{x}) w_i(\mathbf{x}) d\mathbf{x} dt = 0, \quad i = 1, 2, \dots, n$$

- A variety of FE schemes arise from the definition of the weighting functions and of the choice of the shape functions
 - Galerkin: the weighting functions are chosen to be the shape functions (the two functions are then often called basis functions or test functions)
 - Subdomain method: the weighting function is chosen to be unity in the sub-region over which it is applied
 - Collocation Method: the weighting function is chosen to be a Dirac-delta

Finite Elements:

Method of Weigthed Residuals, Cont'd

- Galerkin: $\int_t \int_V R(\mathbf{x}) \phi_i(\mathbf{x}) d\mathbf{x} dt = 0, \quad i = 1, 2, \dots, n$
 - Basis functions formally required to be complete set of functions
 - Can be seen as “residual forced to zero by being orthogonal to all basis functions”

- Subdomain method:

$$\int_t \int_{V_i} R(\mathbf{x}) d\mathbf{x} dt = 0, \quad i = 1, 2, \dots, n$$

- Non-overlapping domains V_i often set to elements
- Easy integration, but not as accurate

- Collocation Method: $\int_t \int_V R(\mathbf{x}) \delta_{x_i}(\mathbf{x}) d\mathbf{x} dt = 0, \quad i = 1, 2, \dots, n$

- Mathematically equivalent to say that each residual vanishes at each collocation points $x_i \Rightarrow$ Accuracy strongly depends on locations x_i .
- Requires no integration.

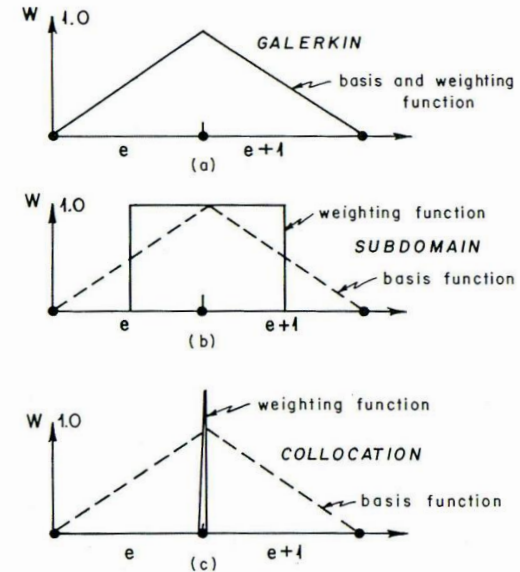


Figure 2.4. Schematic representation of the one-dimensional weighting functions for the Galerkin, subdomain and collocation methods. (It is assumed here that the chapeau function is used as a basis for all methods.)

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